

Elementary Motion Detector in the Female *Drosophila* Optic Lobe: Circuit Structure and Biological Significance from the FAFB

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ABSTRACT

The *Drosophila* Elementary Motion Detector (EMD) implements the Hassenstein-Reichardt correlator for visual motion computation. T4 neurons respond to moving ON (bright edge) and T5 neurons to OFF (dark edge) stimuli, each comprising four subtypes tuned to the four cardinal directions of motion.^[1,2] Silencing both T4 and T5 completely abolishes direction-selective responses in lobula plate tangential cells and eliminates all optomotor behavioral

responses.^[2] The GABAergic hub CT1 spans medulla M10 and lobula Lo1, providing approximately 15 inhibitory synapses per column to T4 and approximately 60 to T5, and is the sole inhibitory columnar input to T5.^[3,4] CT1 receives excitatory input from Tm9 and Tm1 and implements null-direction suppression via a disynaptic mechanism.^[7] Maximum common induced subgraph

(MCIS) search across five *Drosophila* connectome datasets recovers a 45-neuron, 84-directed-edge circuit in FAFB, verified by exhaustive pairwise isomorphism check (2,025 ordered pairs, 0 violations). FlyWire Codex identifies all 45 neurons as this optic lobe EMD circuit. Three neurons are left hemisphere (CT1, cMLLP02, DNc02); 42 are right hemisphere.^[5,6]

CIRCUIT COMPOSITION (FLYWIRE CODEX, FAFB)

LAYER	CELL TYPES	N	NT
Input	Tm9×8, Tm1×2, Tm2×1	11	ACH
Columnar	Mi1×2	2	ACH
ON det. (a-c)	T4a×6, T4b×5, T4c×3	14	ACH
ON det. (d) *	T4d×2	2	GLUT
OFF det. **	T5a×4, T5b×2, T5c×5, T5d×1	12	ACH
Hub (L)	CT1×1	1	GABA
Modulator (R)	0A-AL2i2×1	1	OCT
Modulator (L)	cMLLP02×1	1	ACH
Output (L)***	DNc02×1	1	DA
Total		45	3L / 42R

* T4d: glutamatergic per FlyWire Codex

** One T5c (LO.LOP.3810): no NT annotation in Codex

*** DNc02 also labeled "putative SIFamide" in Codex

Source: FlyWire Codex, FAFB v783^[5,6]

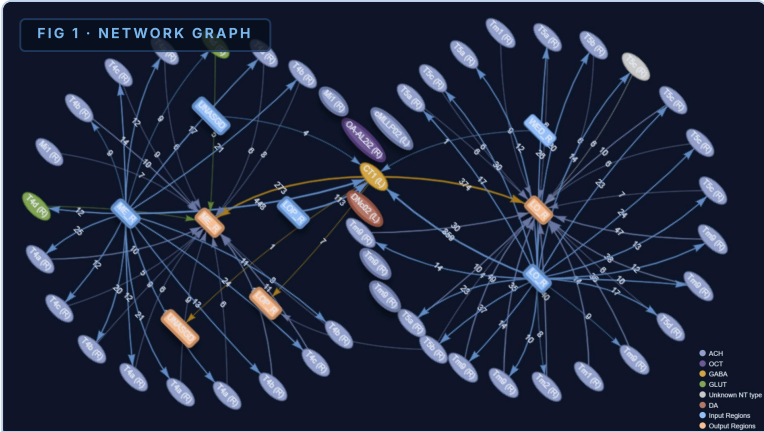


Figure 1. Network graph of the 45-neuron EMD circuit in FAFB (FlyWire Codex). CT1 (L) is the sole GABAergic hub. Edge weights = synapse counts. Colors: ACH (blue), GABA/CT1 (gold), OCT (purple), GLUT (green), DA (brown). Source: FlyWire Codex, FAFB^[5,6].

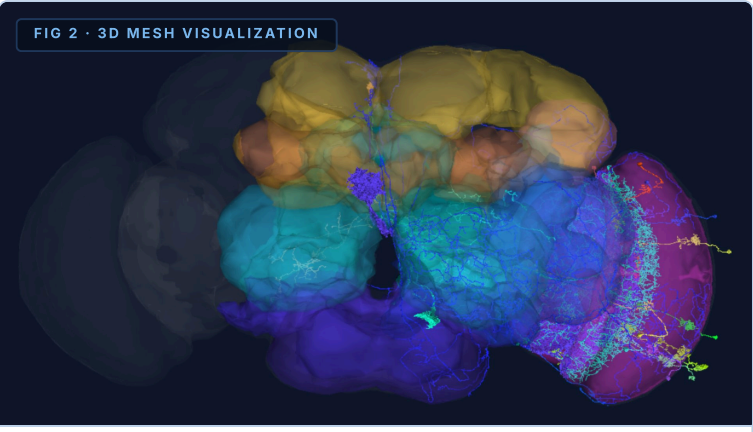


Figure 2. 3D skeleton rendering of all 45 neurons in FAFB (FlyWire Codex). The optic lobe columnar array occupies the right neuropil; the descending DNc02 projection extends toward the central brain and ventral nerve cord. Source: FlyWire Codex, FAFB^[5,6].

CT1 — THE INHIBITORY HUB

- CT1 spans medulla stratum M10 and lobula stratum Lo1 as a single giant GABAergic tangential neuron, forming one synaptic module per column across both neuropils.^[3]
- CT1 provides approximately 15 inhibitory synapses per column to T4 dendrites and approximately 60 per column to T5 dendrites, equal across all four directional subtypes, making it the dominant inhibitory input to the direction-selective layer.^[4]
- CT1 is the sole GABAergic columnar input to T5 dendrites (GAD1-immunopositive). It receives from Tm9 and Tm1 and delivers sign-inverted inhibitory signals onto T5, implementing null-direction suppression via a disynaptic mechanism.^[4,7]
- Three neurons in this circuit are left hemisphere: CT1, cMLLP02, and DNc02. The remaining 42 are right hemisphere. CT1's cross-midline architecture enables pan-columnar inhibition across the entire contralateral optic lobe.^[5,6]

WHAT THIS CIRCUIT COMPUTES

- T4 and T5 are the first direction-selective neurons in the fly visual pathway: T4 responds to ON (bright edge) and T5 to OFF (dark edge) motion, through parallel streams from lamina neurons L1 and L2 respectively.^[1,2]
- Each class comprises four subtypes (a-d) tuned to front-to-back, back-to-front, upward, and downward motion, projecting to corresponding lobula plate layers. All four subtypes of both T4 and T5 are present here, encoding complete 360-degree directional coverage.^[2]
- Direction selectivity arises through preferred-direction enhancement and null-direction suppression on opposing sides of each receptive field, in all four subtypes of both T4 and T5.^[1]
- Silencing both T4 and T5 completely abolishes direction-selective responses in lobula plate tangential cells and eliminates all optomotor behavioral responses.^[2]
- Mi1 is absolutely essential for proper functioning of the ON pathway; blocking Mi1 abolishes T4 responses under all tested conditions.^[1] Tm1, Tm2 and Tm9 relay signals to T5 dendrites in Lo1.^[4]

HYPOTHESIS

CT1's structural centrality reflects its functional role as the sole wide-field GABAergic neuron providing inhibitory input to all direction-selective subtypes.
By receiving excitatory input from Tm9 and Tm1 and delivering sign-inverted GABA signals onto T5, CT1 converts cholinergic transmedullary inputs into null-direction suppression through a disynaptic microcircuit.^[7]
This cross-midline, pan-columnar hub topology creates a maximally distinctive graph-structural signature, explaining why the EMD is recovered as the largest maximum common induced subgraph across connectome datasets without biological labels.